From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

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***Abstract*—Printed rolling-disc notched tubes are cheap tendon-driven continuum robots, but there is no rule for choosing their notches and no live model of their shape that needs no sensor. This report covers both. Part A, submitted to ICRA 2027, maps the notch geometry to bend, stroke and force with an automated CAD-to-FEM pipeline (616 simulations) and three prototypes: the disc diameter sets curvature, the length sets reach, and 42 to 47 % of the tendon stroke is axial compression. Part B, in preparation for IEEE RA-L, is a digital twin that predicts the whole backbone from the servo displacements alone. The module is a beam whose cross-sections come from the CAD; the tendon map is exactly linear and symmetric positive definite, so pull-only tensions solve a strictly convex program, and gravity enters by a fixed point. Instead of learning the shape, we identify the module's compliance law through the exact solver on one and two modules. Unchanged, it predicts four modules (640 mm) to 0.50 mm of FEM, 0.08 % of length, in 1 to 4 ms per pose, where constant curvature errs by 121 mm and a physics-informed network must be retrained.**

***Index Terms*—Continuum robots, tendon-driven, digital twin, finite element method, system identification, scientific machine learning.**

I. Introduction

A tendon-driven continuum robot (TDCR) bends along its whole body when tendons in its wall are pulled, which suits confined spaces [1], [2]. Ours is a printed TPU tube with two orthogonal families of rolling-disc notches (Fig. 1); 160 mm modules stack, and distal tendons run through proximal modules. Two questions drive the work. *Design:* which geometry gives a wanted bend, and what stroke and force does it cost? *Model:* can the whole shape of a stack be predicted in real time from the servo commands, with no sensor on the robot, for any number of modules? The contributions are:

1) SWEEP, an unattended CAD-to-FEM pipeline, and a 616-run study of how notch geometry sets curvature, reach, stroke and tension (Section III; ICRA 2027, second author).
2) A twin built from the CAD's sections with an exact, convex pull-only solver and gravity (Section IV).
3) Identification of the module's law, not its shape, through that solver, and its transfer from one and two modules to four (Sections V, VI).

ME 299 project report, IIT Gandhinagar, November 2026. Code, data and results: https://github.com/om-mpawar/ME299-25110218.

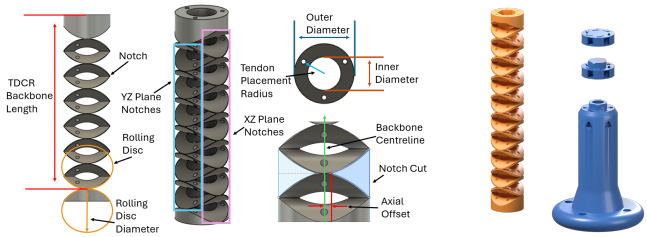


Fig. 1. The module and its design parameters (left): disc diameter D_{RD}, length L, outer and inner diameters D_o, D_i, tendon radius, notch offset. Right: parts designed for the four-module build (TPU module, PLA couplers, base column).

II. Related Work

Constant-curvature (CC) kinematics treat each section as an arc [1]. Camarillo et al. model tendon loads as a moment and an axial force on every section passed, with coupled routing and slack as a complementarity problem [3], [4]; I derived the same pull-only model before finding it, verified the two agree to 3×10^{-6}, and moved the contribution to identification and transfer. Cosserat rods give the same physics by shooting [5], [6]; SOFA with SoftRobots is our FEM ground truth [7]–[9]. Shape estimation uses sensors or learned residuals [10], [11], and physics-informed networks (PINNs) learn one robot's solution [12]–[14]. The closest work calibrates pneumatic modules one at a time [15]; with tendons every module is loaded by those above it, so none can be calibrated alone. Errors are in % of length [16].

III. Part A: How Geometry Governs Compliance

A. Parametric backbone

A notch is cut from a tube of outer radius R_o by two discs of radius $R_d = D_{\mathrm{RD}}/2$ whose centres are D_{RD} apart, so its height and the nominal pitch with a clearance c are

$$h = D_{\mathrm{RD}} - 2\sqrt{R_d^2 - R_o^2}, \qquad p_0 = h + c. \tag{1}$$

On a notched length L with a top margin m_t, the number of notches and the pitch they are spread to are

$$N_n = \left\lfloor \frac{L - m_t - h}{p_0} \right\rfloor + 1, \qquad p = \frac{L - m_t - h}{N_n - 1}, \tag{2}$$

which for $D_{\mathrm{RD}} = 30$ mm and $L = 150$ mm gives $h = 13.4$ mm, ten notches and $p = 14.79$ mm, as in the CAD. The disc

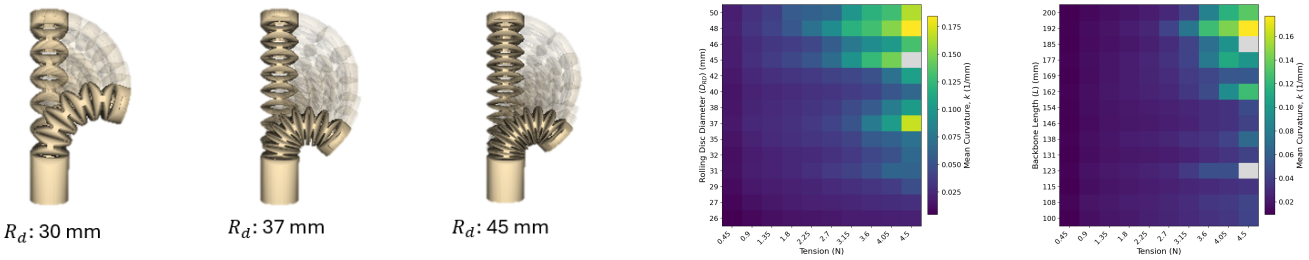


Fig. 2. Part A. Left: the same load on three disc diameters (R_d in the labels is D_{RD}); a larger disc packs more notches and bends further. Middle and right: mean curvature $\bar{\kappa}$ (1/mm) against tension for the disc-diameter and length sweeps; grey cells did not settle. The claims use $T \le 0.9\,\mathrm{N}$.

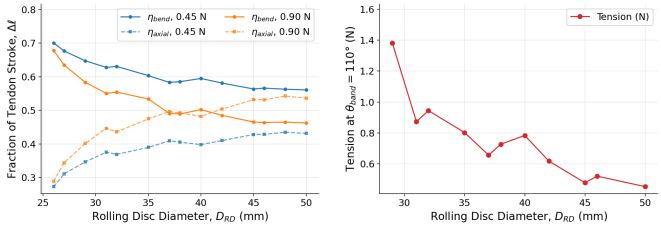
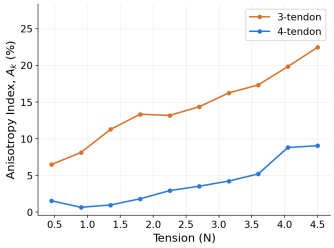


Fig. 3. Stroke budget (5) against D_{RD}. Left: bending and axial fractions at 0.45 and 0.9 N. Right: tension for a 110° bend.



D_{RD} (mm)	centre-line	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

Fig. 4. Left: anisotropy A_κ against tension, three and four tendons ($D_{\mathrm{RD}} = 27\,\mathrm{mm}$). Right: three printed prototypes against simulation, NRMSE in % of length.

diameter therefore sets how many notches fit. Two families at 90° alternate along the tube; each notch leaves two bridges of half-width e.

B. Pipeline and metrics

SWEEP builds each design in CadQuery with one Boolean cut, meshes it with CGAL tetrahedra and pulls one tendon at a time through a load sequence in SOFA, logging the centreline $\{\boldsymbol{p}_i\}$ once the tip has settled. My part was the port to SOFA 25.06, parallel and staged sweeps, a mesh-volume check against the CAD, and per-run settling and self-contact checks. With $\boldsymbol{a} = \boldsymbol{p}_i - \boldsymbol{p}_{i-1}$, $\boldsymbol{b} = \boldsymbol{p}_{i+1} - \boldsymbol{p}_i$ and $\boldsymbol{c} = \boldsymbol{p}_{i+1} - \boldsymbol{p}_{i-1}$, the pointwise (Menger) curvature and its mean are

$$\kappa_i = \frac{2\,\|\boldsymbol{a} \times \boldsymbol{b}\|}{\|\boldsymbol{a}\|\,\|\boldsymbol{b}\|\,\|\boldsymbol{c}\|}, \qquad \bar{\kappa} = \frac{1}{n}\sum_i \kappa_i. \tag{3}$$

The departure from a circular arc is the RMS radial residual to the best-fit (Kåsa) circle $(\boldsymbol{o}, \rho)$, normalised by the arc length ℓ; the anisotropy is the spread of $\bar{\kappa}$ over the tendon directions j at one load:

$$e_{\mathrm{CC}} = \frac{1}{\ell}\Big(\frac{1}{n}\sum_i \big(\|\boldsymbol{p}_i - \boldsymbol{o}\| - \rho\big)^2\Big)^{1/2}, \qquad A_\kappa = \frac{\mathrm{std}_j\,\bar{\kappa}_j}{\mathrm{mean}_j\,\bar{\kappa}_j}. \tag{4}$$

A tendon at offset d that turns the tip by Θ and shortens the body by ΔL needs the stroke

$$\Delta\ell = \Theta d + \Delta L, \qquad \eta_b = \frac{\Theta d}{\Delta\ell}, \quad \eta_a = \frac{\Delta L}{\Delta\ell}. \tag{5}$$

C. Results

The study ran 14 designs × 4 tendons × 11 loads (616 runs, 614 completed), $E = 56\,\mathrm{MPa}$ (measured), with gravity. Above 0.9 N runs stop settling or self-contact, so claims hold for $T \le 0.9\,\mathrm{N}$. Raising D_{RD} from 26 to 50 mm raises the notch count from 4 to 12 and the bend at 0.9 N from 54° to 143°, and e_{CC} about five-fold (Fig. 2). Doubling L from 100 to 200 mm doubles the tip travel, 78 to 157 mm, with $\bar{\kappa}$ changing by under 20 %. The budget (5) closes to $\eta_b + \eta_a = 0.99$ on notched designs (0.72 on a plain tube), and 42 to 47 % of the stroke is compression (Fig. 3); a larger disc lowers the tension for 110° from 0.87 to 0.45 N. Four tendons suit the two notch planes: A_κ falls from 6.2 to 2.6 % in 13 of 14 designs (Fig. 4), with two motors driving antagonistic pairs. Three prototypes loaded with weights agree with simulation in shape to 10 to 19 % of length (Fig. 4); the tip agrees less well, as friction and self-contact are not modelled.

IV. Part B: The Physics of the Twin

Problem. The module is the SWEEP design with $D_{\mathrm{RD}} = 30\,\mathrm{mm}$, $L = 150\,\mathrm{mm}$, $D_o/D_i = 25/14\,\mathrm{mm}$, and three tendons in channels at radius 9.5 mm (sections 1–2 at 0/120/240°, 3–4 at 60/180/300°). Given the servo displacements $\delta\boldsymbol{\ell} \in \mathbb{R}^{3N}$ and gravity, predict the backbone's frames every 5 mm with no sensor on the robot.

A. Section properties

Let z be the material coordinate along the backbone. From the CAD we compute, every 0.02 mm and with the channels removed, the area $A(z)$ and second moments $J_x = \int x^2 dA$, $J_y = \int y^2 dA$ (Fig. 6). The backbone is divided into cells c of $\Delta = 0.5\,\mathrm{mm}$ with compliances

$$a_c = \int_c \frac{dz}{EA}, \qquad C_c = \int_c \mathrm{diag}\Big(\frac{1}{EJ_x}, \frac{1}{EJ_y}\Big)dz,$$
$$t_c = \int_c \frac{2(1+\nu)}{E\,(J_x + J_y)}\,dz, \tag{6}$$

for compression, bending and twist ($E = 56\,\mathrm{MPa}$, $\nu = 0.48$).

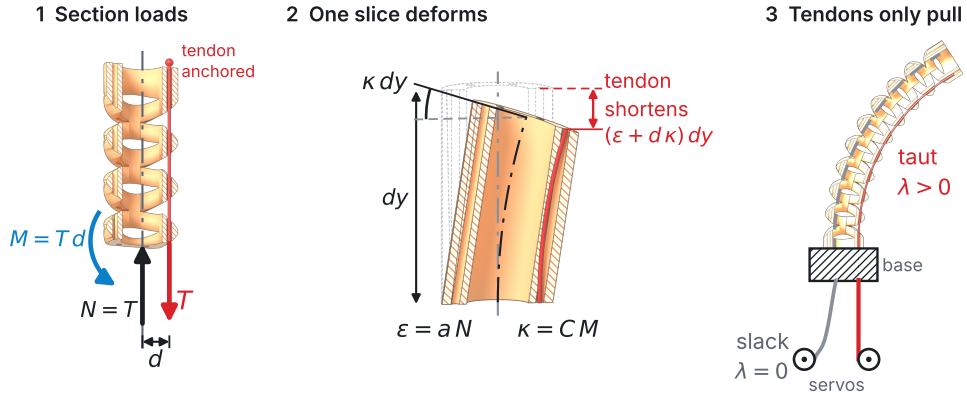


Fig. 5. The physics, drawn on half sections of the CAD. (1) A tendon of tension T at offset d loads every section it passes with $N = T$ and $M = Td$. (2) A slice of height dy shortens by $\varepsilon\, dy$ and bends by $\kappa\, dy$; the tendon's path shortens by $(\varepsilon + d\kappa)\, dy$ (exaggerated). (3) The module bent by the twin: the pulled tendon is taut, the paid-out one slack.

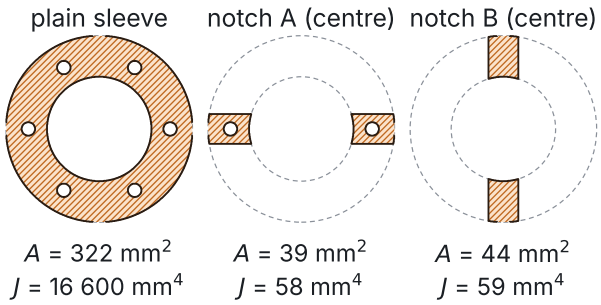


Fig. 6. Cross-sections of the module from the CAD (hatched: material), with the tendon channels removed. A notch keeps two 4 mm bridges: A falls 8-fold and $\min(J_x, J_y)$ 280-fold.

B. Tendon loads and the tendon map

Tendon i, at offset $\boldsymbol{d}_i \in \mathbb{R}^2$ in the cross-section with tension λ_i, applies an axial force λ_i and a moment $\lambda_i \boldsymbol{d}_i$ to every section it passes [3], [5] (Fig. 5). With $\tau_{ic} = 1$ if tendon i passes cell c, the cell's axial force, moment and strains are

$$\begin{aligned} N_c &= \textstyle\sum_i \tau_{ic}\lambda_i, \quad \boldsymbol{m}_c = \textstyle\sum_i \tau_{ic}\lambda_i \boldsymbol{d}_i, \\ \varepsilon_c &= a_c N_c, \qquad \boldsymbol{\kappa}_c = C_c \boldsymbol{m}_c, \end{aligned} \tag{7}$$

where $\boldsymbol{\kappa}_c$ is the integrated curvature towards x and y. Tendon i's path shortens by $\sum_c \tau_{ic}(\varepsilon_c + \boldsymbol{d}_i^\top \boldsymbol{\kappa}_c)$, and the servo also takes up its stretch $c_t L_i \lambda_i$. Since the channels are straight in the material frame, this is linear at any bend:

$$\delta\boldsymbol{\ell} = G\boldsymbol{\lambda}, \qquad G_{ij} = \sum_c \tau_{ic}\tau_{jc}\big(a_c + \boldsymbol{d}_i^\top C_c \boldsymbol{d}_j\big) + c_t L_i\, \delta_{ij}. \tag{8}$$

Proposition 1: G is symmetric positive definite, and $\frac{1}{2}\boldsymbol{\lambda}^\top G\boldsymbol{\lambda}$ is the complementary energy of the backbone and tendons.

Proof: Symmetry is evident from (8). Substituting (7),

$$\tfrac{1}{2}\boldsymbol{\lambda}^\top G\boldsymbol{\lambda} = \tfrac{1}{2}\sum_c \big(a_c N_c^2 + \boldsymbol{m}_c^\top C_c \boldsymbol{m}_c\big) + \tfrac{1}{2}\sum_i c_t L_i \lambda_i^2. \tag{9}$$

It vanishes only if $N_c = 0$ and $\boldsymbol{m}_c = \boldsymbol{0}$ in every cell. In the top module only its own three tendons act, and the vectors $(1, \boldsymbol{d}_i)$ of three channels 120° apart are linearly independent, so their tensions are zero; peeling modules from the tip gives $\boldsymbol{\lambda} = \boldsymbol{0}$. ∎

C. Pull-only tensions

A tendon cannot push. With slack $w_i = (G\boldsymbol{\lambda} - \delta\boldsymbol{\ell})_i$, the tensions satisfy the linear complementarity problem $\boldsymbol{\lambda} \geq \boldsymbol{0}$, $\boldsymbol{w} \geq \boldsymbol{0}$, $\boldsymbol{\lambda}^\top \boldsymbol{w} = 0$, which is exactly the KKT system of

$$\boldsymbol{\lambda}^\star = \arg\min_{\boldsymbol{\lambda}\geq\boldsymbol{0}} \ \tfrac{1}{2}\boldsymbol{\lambda}^\top G\boldsymbol{\lambda} - \delta\boldsymbol{\ell}^\top\boldsymbol{\lambda}. \tag{10}$$

By Proposition 1 it is strictly convex with a unique solution: minimum complementary energy minus the work of the prescribed displacements. With $G = LL^\top$ the objective is $\frac{1}{2}\|L^\top\boldsymbol{\lambda} - L^{-1}\delta\boldsymbol{\ell}\|^2$ up to a constant, so we solve it exactly by non-negative least squares on the Cholesky factor. Slack tendons and the coupling of distal tendons through proximal modules [4] follow from the same G.

D. Kinematics

Each cell turns by the rotation vector $\boldsymbol{\omega}_c = (-\kappa_{c,y},\ \kappa_{c,x},\ \phi_c)$ in its own frame, ϕ_c being the twist. Frames and positions follow by the midpoint rule,

$$\begin{aligned} R_{c+1} &= R_c\, \mathrm{Exp}(\boldsymbol{\omega}_c), \\ \boldsymbol{p}_{c+1} &= \boldsymbol{p}_c + R_c\, \mathrm{Exp}(\tfrac{1}{2}\boldsymbol{\omega}_c)\,(\Delta - \varepsilon_c)\,\boldsymbol{e}_z, \end{aligned} \tag{11}$$

and each joint adds a rigid straight segment. Without gravity E cancels: scaling G scales $\boldsymbol{\lambda}$ inversely.

E. Gravity

Module mass is spread over the cells in proportion to the CAD area, m_c, with weight $\boldsymbol{F}_c = m_c \boldsymbol{g}$. From the current shape, the weight distal to cell c gives, in the cell frame,

$$\boldsymbol{n}_c = R_c^\top \sum_{c'>c} \boldsymbol{F}_{c'}, \qquad \boldsymbol{\mu}_c = R_c^\top \sum_{c'>c} (\boldsymbol{p}_{c'} - \boldsymbol{p}_c) \times \boldsymbol{F}_{c'}, \tag{12}$$

which adds the strains $\boldsymbol{\kappa}_c^g = C_c S \boldsymbol{\mu}_c$, $\varepsilon_c^g = -a_c n_{c,z}$ and the twist $\phi_c = k_T\, t_c\, \mu_{c,z}$, with S picking the bending components and k_T the torsion factor of Section V. The weight also shortens the tendon paths by $h_i = \sum_c \tau_{ic}(\varepsilon_c^g + \boldsymbol{d}_i^\top \boldsymbol{\kappa}_c^g)$, so the tensions are re-solved from (10) with $\delta\boldsymbol{\ell} - \boldsymbol{h}$. Writing $\boldsymbol{x}$ for all gravity strains and Φ for one pass of shape, loads, tensions and strains, the twin iterates

$$\boldsymbol{x}^{k+1} = (1 - \beta_k)\,\boldsymbol{x}^k + \beta_k\, \Phi(\boldsymbol{x}^k), \tag{13}$$

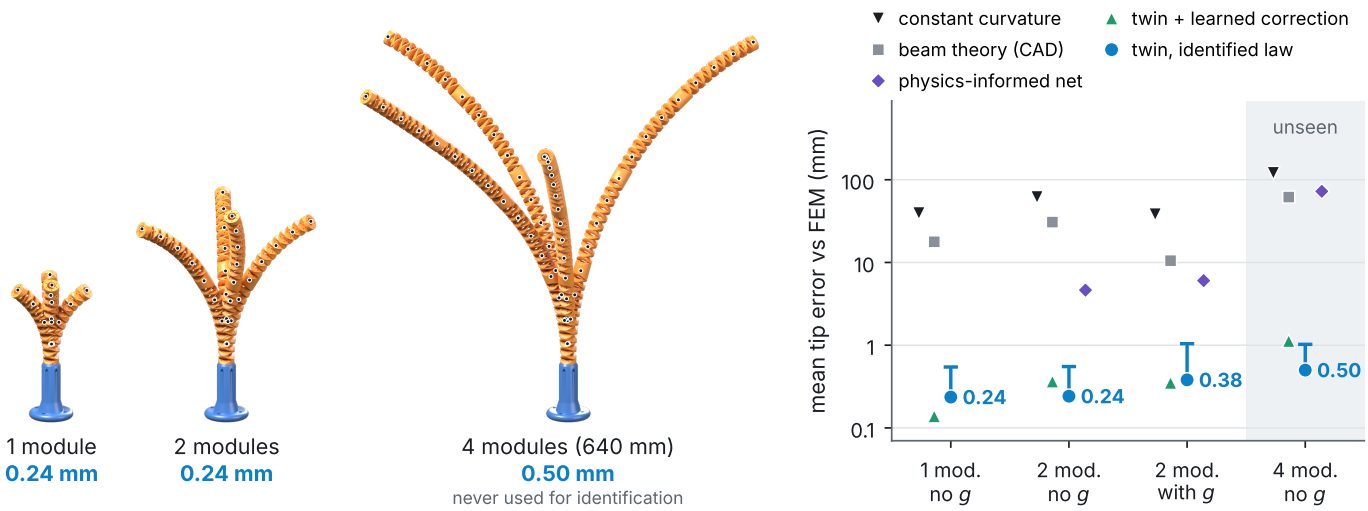


Fig. 7. Transfer to configurations never used for identification. Left: the twin rendered on the CAD for the four most bent held-out poses of 1, 2 and 4 modules, at one scale; dots are the FEM centreline. Right: mean tip error against FEM, bars to the 95th percentile; 60 to 100 held-out poses per set, tip bends up to 75°.

halving β_k (from 0.5, down to 0.05) whenever the update grows, until the change is below 10^{-4} per cell. It is warm-started from the previous call, so following the servos takes a few iterations.

V. Part B: Identifying the Module Law

A. Learn the law, not the solution

Plane sections miss how a short notch deforms (the discs compress like wave springs). Rather than learn shapes, we correct the law (6) and keep the exact solver:

$$a_c(\boldsymbol{\theta}) = a_c^{\mathrm{CAD}} \exp\!\big(s_a \chi_c + \boldsymbol{b}(z_c)^{\top} \boldsymbol{f}_a\big), \tag{14}$$

and likewise for each bending direction, where $\chi_c = 1$ in cells a notch cuts, s is one log-scale per compliance and $\boldsymbol{b}$ a linear spline basis along the module (17 knots). Positive factors keep G positive definite, so the fitted model is still convex and composes to any N. The parameters $\boldsymbol{\theta} = (\boldsymbol{s}, \boldsymbol{f})$ are fitted through the full model to SOFA poses of one and two modules,

$$\min_{\boldsymbol{\theta}} \sum_{\text{sets } q} \frac{1}{P_q K_q} \sum_{p,k} \big\| \boldsymbol{p}_k(\boldsymbol{\theta}; \delta\boldsymbol{\ell}_p, \boldsymbol{g}_q) - \boldsymbol{p}_{pk}^{\mathrm{FEM}} \big\|^2 + \rho \|\boldsymbol{f}\|^2 + \sigma \|D\boldsymbol{f}\|^2, \tag{15}$$

over P_q poses and K_q stations per set, with D the first difference ($\rho = 0.02$, $\sigma = 0.5$), by trust-region least squares with a finite-difference Jacobian. On 675 poses (Fig. 8) the notched cells come out ×20.2 softer axially and ×5.9, ×5.3 in bending; the profile stays within 13 % of constant. A SOFA twist test of one module gives the torsion factor $k_T = 2.45$.

B. What the data can identify

Proposition 2: Scaling every compliance (a_c, C_c, t_c, c_t) by $k > 0$ and every mass by $1/k$ leaves the shape unchanged for every $\delta\boldsymbol{\ell}$ and $\boldsymbol{g}$.

Proof: $G \to kG$ gives $\boldsymbol{\lambda} \to \boldsymbol{\lambda}/k$, so (7) is unchanged; the gravity strains are compliance times weight, also unchanged; hence (13) has the same fixed point. ∎

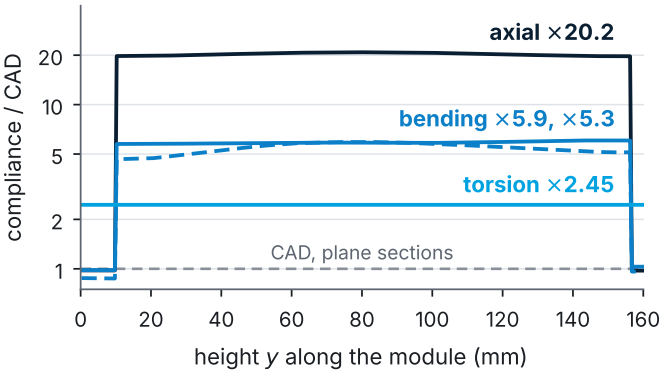


Fig. 8. The identified law (14): each compliance of one module divided by its plane-section value. Four scales carry it; the plain sleeve keeps the CAD value.

Shape data cannot separate softness from weight, so on hardware the modules are weighed, not fitted; without gravity the overall scale is unobservable and a small ridge fixes it.

Proposition 3: For one module with all three tendons taut, rigid tendons and no gravity, the total shortening s and bend $\boldsymbol{\Theta} = \sum_c \boldsymbol{\kappa}_c$ are fixed by $\delta\boldsymbol{\ell}$ alone: $s = \frac{1}{3}\sum_i \delta\ell_i$ and $\boldsymbol{d}_i^{\top}\boldsymbol{\Theta} = \delta\ell_i - s$.

Proof: All tendons pass every cell, so $\delta\ell_i = s + \boldsymbol{d}_i^{\top}\boldsymbol{\Theta}$; channels 120° apart give $\sum_i \boldsymbol{d}_i = \boldsymbol{0}$. ∎

Taut one-module poses therefore carry no information on the compliances. Our FEM poses each have a slack tendon, and then one module does fix the axial-to-bending ratio (Jacobian singular values 622, 545 and 0.41, the weak direction being the overall scale, as Proposition 2 predicts); two modules, where a distal pull divides between modules, make it robust.

Proposition 4: For N identical modules, the pretension giving every tendon the same tension pulls section s by $p_s = p_1 \sum_{j \le s}(N - j + 1)/N$: 1 : 1.75 : 2.25 : 2.5 for four modules.

Proof: With $\boldsymbol{\lambda} = \lambda\boldsymbol{1}$ the bending terms of (8) cancel in

TABLE I
MEAN TIP ERROR AGAINST FEM ON HELD-OUT POSES (MM)

Modules, gravity	1, no	2, no	2, yes	4, no*
Constant curvature	39.5	61.8	38.3	120.9
Beam theory from CAD	17.8	30.8	10.5	61.7
PINN (forward)	–	4.6	6.0	72.5†
Twin + GP correction	0.14	0.36	0.35	1.12
Identified law (mean)	**0.24**	**0.24**	**0.38**	**0.50**
95th percentile	0.54	0.55	1.04	1.02

*never used for fitting. †retrained on four-module physics.

each section ($\sum_{i\in s} \boldsymbol{d}_i = \mathbf{0}$), and module j carries the axial force of the $3(N-j+1)$ tendons passing it. ∎

C. Few-shot hardware calibration

Five numbers are fitted on about 15 motion-capture poses of two modules: log-scales on the axial and two bending compliances, the tendon compliance c_t, and a moment-arm offset Δd (a wire bears nearer the axis than SOFA's cable), minimising $\sum_{p,k} \|\boldsymbol{p}_k - \hat{\boldsymbol{p}}_k\|^2 + \ell^2 \|\mathrm{Log}(R_k^\top \hat{R}_k)\|^2$ with $\ell = 100\,\mathrm{mm}$. On synthetic data with 0.3 mm noise this recovers the bending scale to 0.003 and Δd to 0.02 mm, while a free mass drifts by tens of percent (Proposition 2).

VI. Results

Table I and Fig. 7 compare five models on held-out FEM poses. The CC baseline puts one arc on each module, $\delta\ell_i = r\,\ell_t \sum_{m\le s(i)} \kappa_m \cos(\psi_i - \alpha_m)$, solved for (κ_m, α_m) by least squares; ignoring compression, it over-bends 1.7-fold. A learned (GP) correction helps where trained but doubles the error on four modules, and PINNs must be retrained for four. The identified law stays at 0.24 to 0.50 mm in every set, including four modules (640 mm, 0.08 % of length) never used for fitting; its scales alone, fitted on two-module data, give 0.67 mm on four. A pose takes a median of 1.0, 1.9 and 4.0 ms for 1, 2 and 4 modules with gravity (slowest call 26 ms, laptop CPU), against 357 ms for a Cosserat rod. The solver matches closed-form beam cases to under 1 µm and an independent Cosserat rod to 0.004 mm; 132 automated tests and one command regenerate every number. All errors are against FEM, whose coarse and fine meshes differ by about 0.6 mm at the tip (earlier robot), the scale of the twin's error.

VII. Future Work: Towards Sensor-Free Cavity Insertion

(1) Hardware validation (now): calibrate the five numbers on 15 poses of two modules and test on four, passing if the mean tip error is at most 2 % of length (13 mm), a bar set before any data. *(2) Contact in the twin:* a wall is one-sided like a tendon; contact forces $f_k \ge 0$ with gaps $g_k(\boldsymbol{x}) \ge 0$ and $f_k g_k = 0$ form a second complementarity problem of the same kind as (10), solved inside (13), to be checked against FEM and motion capture with a wall. *(3) Cavity insertion:* feed the stack through an opening while the twin chooses the servo commands, passing if it reaches a target inside a printed cavity. If residuals grow with bend, a convex notch energy replaces the linear law.

VIII. Conclusion and Learning Outcomes

The notch geometry gives separate control of curvature and reach, at the price of a stroke budget in which compression takes almost half. A module law identified inside an exact, convex model of the CAD transfers from one and two modules to four to 0.08 % of length in milliseconds, where constant curvature and learned solutions do not. The project taught continuum-robot mechanics, FEM in SOFA, convex optimisation with unilateral constraints, system identification and scientific machine learning, and a method: test simulators on simple cases (our cables could push), search the literature before claiming novelty, fix pass criteria before taking data, and make every number reproducible.

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